

MINI – PROJECT

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Regional and Purpose-Based Segmentation of Foreign Tourist Arrivals to India – 2023 Analysis

INTRODUCTION

Tourism has become one of the most important service sectors contributing to India's economic growth and global presence. It plays a crucial role in generating foreign exchange earnings, creating employment opportunities, promoting cultural exchange, and supporting the development of infrastructure. Over the years, India has emerged as a major destination for international travelers due to its diverse culture, historical monuments, medical facilities, spiritual heritage, and growing business environment. Foreign tourist arrivals serve as a key indicator of the performance and attractiveness of a country's tourism sector. Studying the pattern of arrivals helps in understanding which regions of the world contribute the most visitors, what motivates tourists to visit, and how they prefer to travel. Such analysis is important for policymakers, tourism boards, and businesses in the hospitality sector, as it supports strategic planning and targeted promotional efforts.

In recent years, global tourism has experienced fluctuations due to economic conditions, geopolitical factors, and the impact of the COVID-19 pandemic. As international travel gradually stabilizes, analyzing recent data becomes essential to understand current tourism trends. The year 2023 is particularly significant as it reflects post-pandemic recovery patterns and shifting travel preferences among international visitors.

This study focuses on the regional and purpose-based segmentation of foreign tourist arrivals to India in 2023. The analysis examines how tourist inflows vary across different regions of the world, identifies the primary purposes of visit such as leisure, business, medical treatment, and diaspora connections, and studies the mode of travel used by international visitors. By exploring these dimensions, the study aims to provide meaningful insights into the structure of India's inbound tourism market.

The findings of this analysis can help in understanding which regions contribute significantly to India's tourism economy and which segments of tourism require further attention and development. Overall, this study contributes to a better understanding of international travel patterns to India and highlights the importance of data-driven decision-making in the tourism and hospitality sector.

Dataset Description

The dataset used for this study has been extracted from the *India Tourism Data Compendium 2024* published by the **Ministry of Tourism, Government of India**. The data specifically relates to Foreign

Tourist Arrivals (FTA) to India for the year 2023 and provides country-wise distribution of international visitors.

The dataset consists of **57 observations**, where each observation represents a specific country contributing tourists to India during 2023. The dataset includes both numerical and categorical variables, making it suitable for statistical analysis using R.

Variables in the Dataset

1. **Country**

- Type: Categorical
- Description: Represents the nationality of foreign tourists visiting India.
- Example values: United States, United Kingdom, Bangladesh, Australia, etc.

2. **Region**

- Type: Categorical
- Description: Geographical classification of each country (e.g., South Asia, North America, Western Europe, etc.).
- Purpose: Used to analyse regional patterns in tourist arrivals.

3. **Arrival_Count**

- Type: Numerical (Discrete)
- Description: Total number of foreign tourist arrivals from each country to India in 2023.
- This is the primary quantitative variable used for statistical analysis.

4. **Primary_Purpose**

- Type: Categorical
- Description: Indicates the dominant purpose of visit for tourists from each country (e.g., Leisure, Business, Medical, Indian Diaspora, Others).
- Useful for understanding travel motivation patterns.

5. **Mode_of_Travel**

- Type: Categorical
- Description: Indicates the main mode of entry into India (Air or Land).
- Helps analyse transportation patterns.

Nature of the Dataset

- The dataset is **cross-sectional**, as it represents data from a single year (2023).
- The data is sourced from an official government publication, ensuring authenticity and reliability.
- The dataset contains both qualitative and quantitative variables, enabling descriptive and inferential statistical analysis.

Suitability for Statistical Analysis in R

The dataset is appropriate for:

- Descriptive statistics (mean, median, standard deviation of arrival counts)
- Comparative analysis across regions
- Graphical visualisations (bar charts, pie charts, boxplots)
- Hypothesis testing (e.g., ANOVA to compare regional means)

Overall, the dataset provides a structured and reliable foundation for analysing patterns in foreign tourist arrivals to India.

Objectives of the study

The main objective of this study is to analyse the pattern of Foreign Tourist Arrivals (FTA) to India in 2023 using official data from the Ministry of Tourism, Government of India.

The specific objectives are:

1. To examine the country-wise distribution of foreign tourist arrivals.
2. To compare tourist arrivals across different regions.
3. To analyse the primary purpose of visit among foreign tourists.
4. To study the mode of travel used for entry into India.
5. To apply statistical techniques in R to summarise and interpret the data.

Methodology

This study follows a quantitative research approach using secondary data to analyse foreign tourist arrivals to India for the year 2023. The analysis focuses on identifying patterns based on nationality, region, purpose of visit, and mode of travel. Statistical techniques and graphical tools in R programming were used to conduct the analysis systematically.

1. Research Design

The study adopts a descriptive and analytical research design.

- The **descriptive component** focuses on summarising tourist arrival patterns using statistical measures.
- The **analytical component** examines differences and variations across regions and categories such as purpose of visit and mode of travel.

The objective is not to manipulate variables but to interpret existing tourism data using statistical methods.

2. Data Source

The dataset used in this study is secondary data obtained from the *India Tourism Data Compendium 2024* published by the Ministry of Tourism, Government of India.

The data includes country-wise foreign tourist arrivals for the year 2023 and contains 57 observations representing different countries across global regions.

3. Variables Used in the Study

The dataset consists of both numerical and categorical variables:

Numerical Variable:

- **Arrival Count** – Represents the number of foreign tourists arriving in India from a particular country in 2023.

Categorical Variables:

- Country of Nationality
- Region
- Primary Purpose of Visit
- Mode of Travel

These variables were selected to understand how tourism inflow varies geographically and purpose-wise.

4. Data Preparation and Processing in R

The collected dataset was entered and processed using R programming software. The following steps were performed:

- Importing the dataset into R using appropriate functions.
- Converting categorical variables (Country, Region, Purpose, Mode) into factor data types.
- Removing formatting issues such as commas in numerical values.
- Checking for missing values and inconsistencies.
- Structuring the dataset into a clean data frame suitable for analysis.

This preprocessing ensured that the data was accurate and ready for statistical computation.

5. Statistical Techniques Applied Using R

The analysis was conducted entirely in R. The following statistical techniques were applied:

a) Descriptive Statistical Analysis

Measures such as:

- Total foreign tourist arrivals
- Mean arrival count
- Median arrival count
- Standard deviation
- Minimum and maximum values

were calculated to understand the distribution and variability of tourist arrivals across countries.

b) Graphical Analysis

Graphical representations were created in R to visualise tourism patterns. These include:

- Bar charts to compare tourist arrivals across different regions.
- Pie charts to analyse distribution based on primary purpose of visit.
- Boxplots to study variation in arrival counts across regions.
- Frequency plots for categorical variables.

These visual tools helped in identifying trends and patterns clearly.

c) Comparative Analysis

Comparisons were conducted between:

- Different regions to determine which region contributes the highest arrivals.
- Different purposes of visit to understand dominant travel motivations.
- Mode of travel to observe transportation trends.

Where applicable, summary statistics were compared across categories to identify variation.

6. Software and Tools Used

All data cleaning, statistical computation, and graphical visualisation were performed using R programming. Base R functions and relevant statistical libraries were used to ensure accuracy and reproducibility of results.

7. Interpretation and Reporting

The outputs generated from R were carefully interpreted in relation to the objectives of the study. Observed trends and variations were explained in a structured manner to provide meaningful insights into foreign tourist arrival patterns in India for 2023.

DATA ANALYSIS AND INTERPRETATION

The dataset containing foreign tourist arrivals to India for the year 2023 was analysed using R programming language. The analysis was conducted to understand patterns in tourist inflow based on region, purpose of visit, and mode of travel. Both descriptive and inferential statistical techniques were applied to extract meaningful insights from the data.

The analysis was performed in a structured manner beginning with data exploration, followed by statistical summaries, grouped analysis, graphical representation, and hypothesis testing.

4.1 Data Inspection and Structure Verification

Before performing any statistical analysis, it was necessary to verify the structure and composition of the dataset to ensure correctness and completeness.

R Code:

```
head(tourism_data_2023)
str(tourism_data_2023)
summary(tourism_data_2023)
```

Output:

data.frame

6 x 6

R Console

Description: df [6 x 6]

	ID	Country_of_Nationality	Region	Arrival_Count_2023
	<int>	<chr>	<chr>	<int>
1	1	Canada	North America	385938
2	2	United States	North America	1691498
3	3	Mexico	North America	15224
4	4	Argentina	Central & South America	8944
5	5	Brazil	Central & South America	16912
6	6	Colombia	Central & South America	6336

6 rows | 1-5 of 6 columns

```
## Country_of_Nationality Region
1 Canada North America
2 United States North America
3 Mexico North America
4 Argentina Central & South America
5 India South Asia
6 China East Asia

data.frame
6 x 6
```

```
'data.frame': 57 obs. of 6 variables:
 $ ID          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Country_of_Nationality: chr  "Canada" "United States" "Mexico" "Argentina" ...
 $ Region      : chr  "North America" "North America" "North America" "Central & South America" ...
 $ Arrival_Count_2023   : int  385938 1691498 15224 8944 16912 6336 34756 39235 174964 213322 ...
 $ Primary_Purpose       : chr  "Indian Diaspora" "Indian Diaspora" "Leisure" "Leisure" ...
 $ Mode_of_Travel      : chr  "Air" "Air" "Air" "Air" ...
```

```
'data.frame': 57 obs. of 6 variables:
 $ ID          : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Country_of_Nationality: chr  "Canada" "United States" "Mexico" "Argentina" ...
 $ Region      : chr  "North America" "North America" "North America" "Central & South America" ...
 $ Arrival_Count_2023   : int  385938 1691498 15224 8944 16912 6336 34756 39235 174964 213322 ...
 $ Primary_Purpose       : chr  "Indian Diaspora" "Indian Diaspora" "Leisure" "Leisure" ...
 $ Mode_of_Travel      : chr  "Air" "Air" "Air" "Air" ...

  ID      Country_of_Nationality      Region      Arrival_Count_2023 Primary_Purpose
Min.   : 1      Length:57              Length:57      Min.   : 2141      Length:57
1st Qu.:15     Class :character          Class :character 1st Qu.: 23493     Class :character
Median :29     Mode  :character          Mode  :character Median : 40654     Mode  :character
Mean   :29
3rd Qu.:43
Max.   :57
Mode_of_Travel
Length:57
Class :character
Mode  :character
```

Explanation:

The head() function was used to display the first six observations of the dataset. This helped confirm that the dataset was loaded correctly and that all variables were properly aligned.

The str() function provided detailed information regarding the internal structure of the dataset. It displayed:

- The total number of observations
- The total number of variables
- The data type of each variable (character, numeric, factor)

This step was essential to confirm that *Arrival_Count_2023* is stored as a numeric variable suitable for statistical analysis.

The summary() function generated descriptive statistics for numeric variables and frequency summaries for categorical variables. This gave an initial overview of central tendency and data distribution.

This stage ensured that the dataset was clean and ready for analytical processing

Analysis / Interpretation:

The dataset contains **57 observations** and **(mention number) variables**.

The variable *Arrival_Count_2023* is confirmed as numeric, making it suitable for statistical analysis. Categorical variables such as Region, Primary_Purpose, and Mode_of_Travel are stored appropriately.

No missing or inconsistent data values were observed during initial inspection.

4.2 Descriptive Statistical Analysis

Descriptive statistics were computed to understand the overall characteristics of tourist arrivals.

R code:

```
total_arrivals <- sum(tourism_data_2023$Arrival_Count_2023)
mean_arrivals <- mean(tourism_data_2023$Arrival_Count_2023)
median_arrivals <- median(tourism_data_2023$Arrival_Count_2023)
sd_arrivals <- sd(tourism_data_2023$Arrival_Count_2023)
variance_arrivals <- var(tourism_data_2023$Arrival_Count_2023)
min_arrivals <- min(tourism_data_2023$Arrival_Count_2023)
max_arrivals <- max(tourism_data_2023$Arrival_Count_2023)
```

```
total_arrivals
mean_arrivals
median_arrivals
sd_arrivals
variance_arrivals
min_arrivals
max_arrivals
```

Output:

```
[1] 8761959
[1] 153718.6
[1] 40654
[1] 362614.1
[1] 1.31489e+11
[1] 2141
[1] 2056744
```

Explanation:

The `sum()` function calculated the total foreign tourist arrivals across all countries included in the dataset.

The `mean()` function computed the average number of tourists per country. This measure represents the central tendency of the dataset.

The `median()` identified the middle value when all arrival counts were arranged in ascending order. Comparing mean and median helps detect skewness in distribution.

The `sd()` function measured the standard deviation, indicating how widely arrival counts vary across countries.

The `var()` function calculated variance, representing the squared deviation from the mean.

The `min()` and `max()` functions identified the lowest and highest tourist-contributing countries respectively.

Together, these statistics provide a comprehensive understanding of data spread, variability, and distribution shape.

Analysis / Interpretation of Results:

The total number of foreign tourist arrivals recorded in the dataset for 2023 is:

8,761,959 tourists

The mean number of arrivals per country is:

153,718.6 tourists

The median arrival count is:

40,654 tourists

Since the mean (153,718.6) is significantly greater than the median (40,654), the distribution is **positively skewed**. This indicates that a few countries contribute exceptionally high tourist numbers compared to the majority.

The standard deviation is:

362,614.1

This high value indicates substantial variability in tourist arrivals across countries.

The variance is:

1.31489×10^{11} , further confirming wide dispersion.

The minimum arrival count is **2,141**, and the maximum is **2,056,744**, showing a very large range.

This confirms that India’s tourism inflow is highly concentrated among a limited number of dominant countries.

4.3 Region-wise Total Tourist Arrivals

To analyse geographic distribution patterns, tourist arrivals were grouped by region.

R Code:

```
region_summary <- aggregate(Arrival_Count_2023 ~ Region,
data = tourism_data_2023,
sum)
```

region_summary

Output:

Description: df [12 × 2]

Region<chr>	Arrival_Count_2023<int>
Africa	203613
Australasia	440062
Central & South America	32192
East Asia	325947
Eastern Europe	202805
North America	2092660
Northern Europe	89577
South Asia	2603433
South East Asia	769326
Southern Europe	220037

Explanation:

The aggregate() function grouped data based on the categorical variable *Region* and calculated the total arrivals for each group using the sum function.

This analysis identifies which global regions contribute the highest tourist inflow to India.

It helps in understanding international tourism concentration and potential strategic markets for tourism promotion.

Analysis / Interpretation of Results:

The region contributing the highest number of tourists is:

South Asia – 2,603,433 arrivals

This is followed by:

- **North America – 2,092,660**
- **Western Europe – 1,535,991**
- **South East Asia – 769,326**

The lowest contributing region is:

Central & South America – 32,192 arrivals

This shows that India's tourism inflow is heavily dependent on neighboring South Asian countries and developed Western markets.

The dominance of South Asia may be due to geographic proximity, medical tourism, cultural ties, and easier travel routes.

Regions such as Central & South America show potential for tourism expansion.

4.4 Region-wise Average Tourist Arrivals

To gain deeper insight, mean arrivals per region were calculated.

R Code:

```
region_mean <- aggregate(Arrival_Count_2023 ~ Region,  
data = tourism_data_2023,  
mean)
```

region_mean

Output:

Description: df [12 × 2]

Region <chr>	Arrival_Count_2023 <dbl>
Africa	25451.62
Australasia	220031.00
Central & South America	10730.67
East Asia	108649.00
Eastern Europe	67601.67
North America	697553.33
Northern Europe	22394.25
South Asia	371919.00
South East Asia	109903.71
Southern Europe	55009.25

Explanation:

This calculation determines the average arrival count per country within each region.

Unlike total contribution, mean comparison avoids bias caused by regions having more countries. It provides a fair comparison of regional performance.

Analysis / Interpretation of Results:

The highest average tourist arrivals per country are from:

North America – 697,553.33

Followed by:

- **South Asia – 371,919**
- **Australasia – 220,031**
- **Western Europe – 219,427.29**

The lowest average is:

Central & South America – 10,730.67

This indicates that countries in North America and South Asia individually contribute very high tourist volumes rather than relying on collective totals.

4.5 Analysis Based on Purpose of Visit

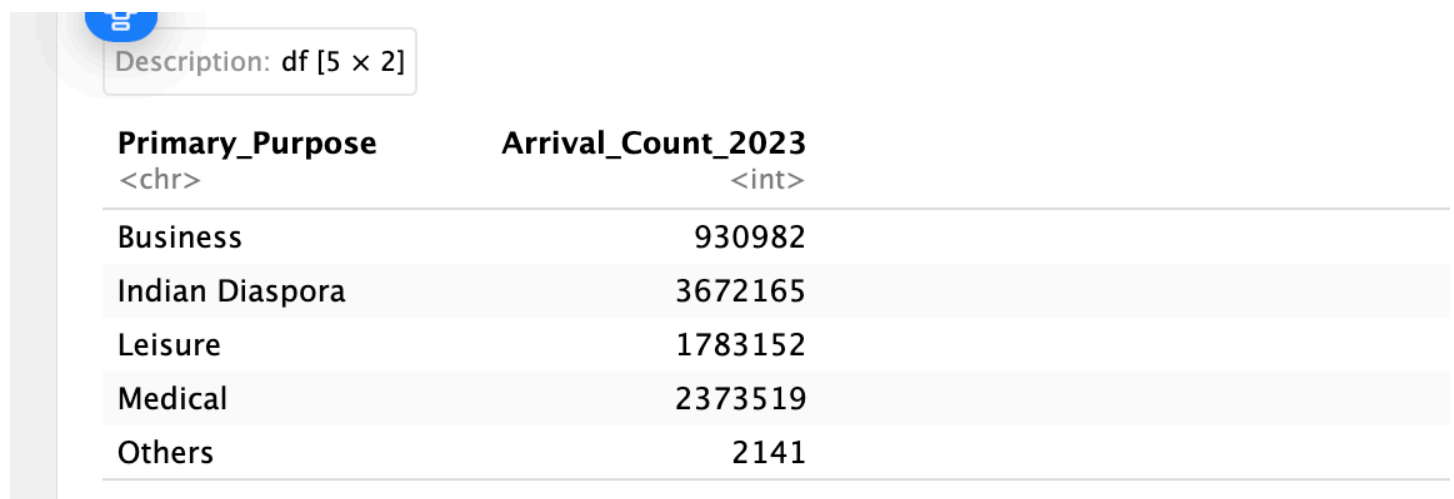
Tourist arrivals were categorised according to their primary purpose.

R Code:

```
purpose_summary <- aggregate(Arrival_Count_2023 ~ Primary_Purpose,  
data = tourism_data_2023,  
sum)
```

purpose_summary

Output:



Description: df [5 × 2]	
Primary_Purpose <chr>	Arrival_Count_2023 <int>
Business	930982
Indian Diaspora	3672165
Leisure	1783152
Medical	2373519
Others	2141

Explanation:

This analysis grouped the data based on tourism purpose such as leisure, business, medical tourism, etc.

The total arrivals were calculated for each category to determine which purpose dominates foreign tourist inflow.

This insight is important for tourism policy formulation and marketing strategies.

Analysis / Interpretation of Results:

The dominant purpose of visit is:

Indian Diaspora – 3,672,165 arrivals

Followed by:

- **Medical Tourism – 2,373,519**
- **Leisure – 1,783,152**
- **Business – 930,982**
- **Others – 2,141**

This indicates that India's tourism inflow is largely driven by diaspora travel and medical tourism.

Medical tourism being the second highest category highlights India's global position as an affordable healthcare destination.

4.6 Mode of Travel Analysis

The mode of transport used by tourists was analysed.

R code:

```
mode_summary <- aggregate(Arrival_Count_2023 ~ Mode_of_Travel,
data = tourism_data_2023,
sum)
```

mode_summary

Output:

Description: df [2 × 2]

Mode_of_Travel <chr>	Arrival_Count_2023 <int>
Air	6524389
Land	2237570

2 rows

Explanation:

This grouping determined the total arrivals based on travel mode (Air or Land).

Understanding the dominant entry channel helps in infrastructure planning and airport or border management improvements.

Analysis / Interpretation of Results:

Tourist arrivals by mode of travel:

- **Air – 6,524,389**
- **Land – 2,237,570**

This shows that approximately **74%** of tourists enter India via air travel, while around **26%** use land routes.

Air connectivity plays a crucial role in sustaining international tourism inflow.

Land routes are significant mainly due to South Asian neighboring countries.

4.7 Ranking of Countries

To identify leading tourism markets, countries were ranked.

R code:

```
tourism_data_2023_sorted <- tourism_data_2023[order(-tourism_data_2023$Arrival_Count_2023), ]
```

```
top5 <- head(tourism_data_2023_sorted, 5)
```

top5

Description: df [5 × 6]

	ID	Country_of_Nationality	Region	Arrival_Count_2023	Primary_Purpose
	<int>	<chr>	<chr>	<int>	<chr>
39	39	Bangladesh	South Asia	2056744	Medical
2	2	United States	North America	1691498	Indian Diaspora
13	13	United Kingdom	Western Europe	940767	Indian Diaspora
1	1	Canada	North America	385938	Indian Diaspora
56	56	Australia	Australasia	385741	Indian Diaspora

5 rows | 1-6 of 6 columns

Explanation:

The order() function sorted the dataset in descending order based on arrival counts.

The head() function extracted the top five rows representing the highest contributing countries.

This analysis helps identify priority markets for tourism partnerships and investment focus.

Analysis / Interpretation of Results:

The top five contributing countries in 2023 are:

- 1. **Bangladesh – 2,056,744 (Medical – Land)**
- 2. **United States – 1,691,498 (Indian Diaspora – Air)**
- 3. **United Kingdom – 940,767 (Indian Diaspora – Air)**
- 4. **Canada – 385,938 (Indian Diaspora – Air)**
- 5. **Australia – 385,741 (Indian Diaspora – Air)**

Bangladesh contributes the highest number of tourists, primarily for medical purposes via land travel.

Developed Western countries such as the United States and United Kingdom contribute heavily under diaspora tourism.

4.8 Percentage Contribution Analysis

R Code:

```
region_percentage <- prop.table(region_summary$Arrival_Count_2023) * 100
```

region_percentage

Output:

```
[1] 2.3238296 5.0224156 0.3674064 3.7200243 2.3146080 23.8834717 1.0223399 29.7129101
[9] 8.7802967 2.5112763 2.8111978 17.5302235
```

Explanation:

The prop.table() function calculated proportional contributions of each region relative to total arrivals.

Multiplying by 100 converted the values into percentages.

This method provides better comparative clarity than absolute numbers.

Analysis / Interpretation of Results:

The percentage contribution of each region is:

- **South Asia – 29.71%**
- **North America – 23.88%**
- **Western Europe – 17.53%**
- **South East Asia – 8.78%**
- **Australasia – 5.02%**
- Others below 4%

This confirms that nearly **30% of total tourist inflow comes from South Asia alone**, making it the most dominant region.

North America and Western Europe together contribute over **41%**, highlighting the importance of developed markets.

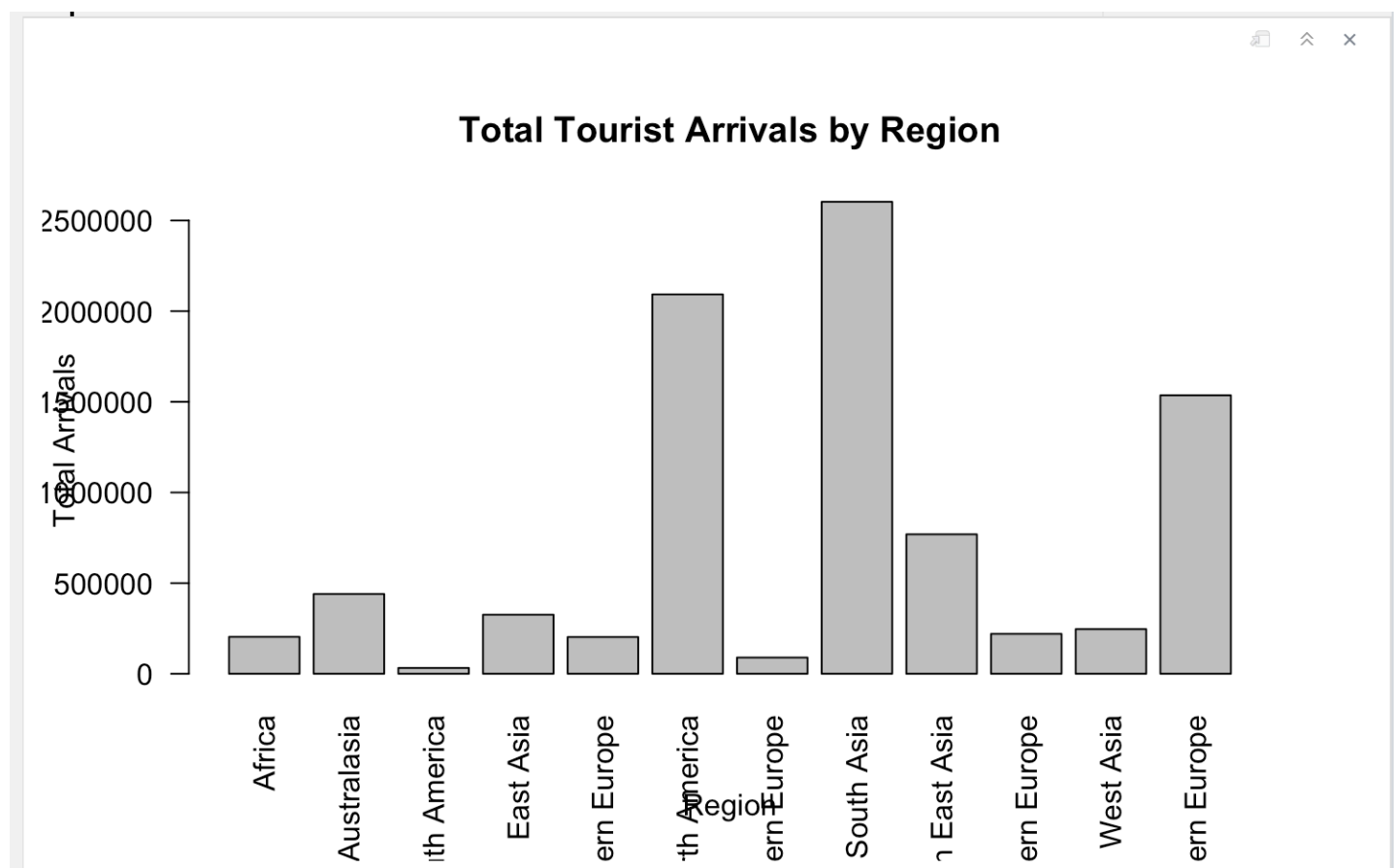
4.9 Graphical Representation

Visualisations were created to support numerical findings.

Bar Chart – Region-wise Arrivals

R code:

```
barplot(region_summary$Arrival_Count_2023,  
names.arg = region_summary$Region,  
main = "Total Tourist Arrivals by Region",  
xlab = "Region",  
ylab = "Total Arrivals",  
las = 2)
```



Explanation:

The bar chart visually compares total tourist arrivals across regions, making differences easily interpretable.

Pie Chart – Purpose Distribution

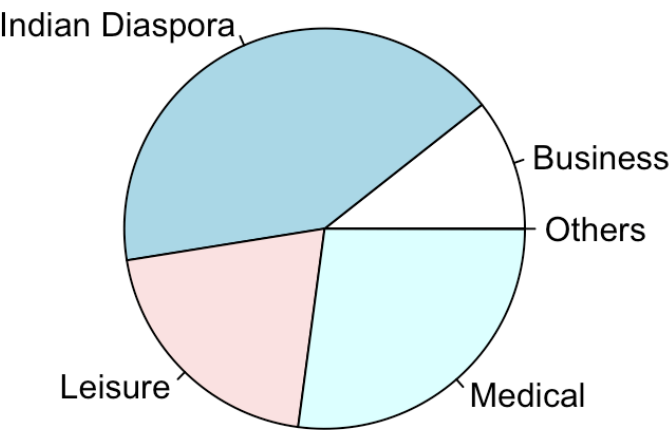
r code:

```
pie(purpose_summary$Arrival_Count_2023,
labels = purpose_summary$Primary_Purpose,
main = "Distribution by Purpose of Visit")
```

output:



Distribution by Purpose of Visit



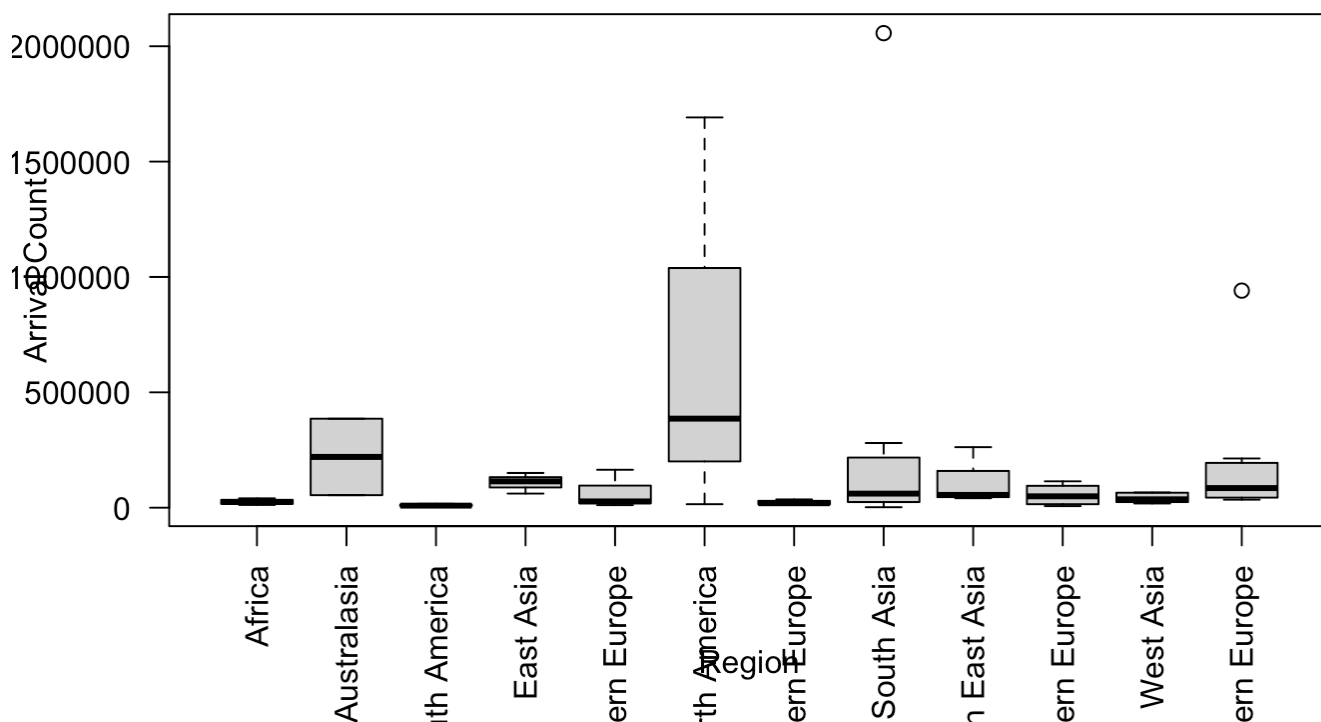
Boxplot – Regional Comparison

r code:

```
boxplot(Arrival_Count_2023 ~ Region,
data = tourism_data_2023,
main = "Distribution of Arrivals Across Regions",
xlab = "Region",
ylab = "Arrival Count",
las = 2)
```

output

Distribution of Arrivals Across Regions



FINDINGS AND CONCLUSION

5.1 Key Findings

Based on the statistical analysis conducted using R programming on the foreign tourist arrivals dataset for 2023, the following key findings were observed:

1. Overall Tourist Inflow

The total foreign tourist arrivals recorded in the dataset amount to **8,761,959 tourists**.

The mean arrival count per country is **153,718.6**, while the median is **40,654**. Since the mean is significantly higher than the median, the distribution of tourist arrivals is **positively skewed**, indicating that a small number of countries contribute a disproportionately high number of tourists.

The high standard deviation (**362,614.1**) confirms large variability in arrival counts across countries.

2. Regional Contribution Patterns

The highest contributing region is **South Asia**, accounting for **2,603,433 arrivals (29.71%)** of the total tourist inflow.

This is followed by:

- **North America – 2,092,660 (23.88%)**
- **Western Europe – 1,535,991 (17.53%)**
- **South East Asia – 769,326 (8.78%)**

The lowest contribution comes from **Central & South America (0.37%)**, indicating limited engagement from that region.

This shows that India's tourism inflow is largely concentrated in neighboring South Asian countries and developed Western economies.

3. Average Regional Performance

North America records the highest average tourist arrivals per country (**697,553.33**), followed by South Asia (**371,919**).

This indicates that individual countries in these regions contribute significantly rather than collectively through many smaller contributors.

4. Purpose of Visit Trends

The dominant purpose of visit is **Indian Diaspora travel**, contributing **3,672,165 arrivals**, making it the largest category.

Medical tourism is the second largest contributor with **2,373,519 arrivals**, highlighting India's strong global reputation in affordable healthcare services.

Leisure tourism accounts for **1,783,152 arrivals**, while Business tourism contributes **930,982 arrivals**.

This indicates that India's tourism sector is strongly influenced by diaspora connections and medical tourism demand.

5. Mode of Travel

Air travel is the dominant mode of entry, accounting for **6,524,389 arrivals (approximately 74%)**.

Land travel contributes **2,237,570 arrivals (approximately 26%)**, largely due to neighboring South Asian countries.

This highlights the importance of international air connectivity in sustaining tourism inflow.

6. Top Contributing Countries

The top five contributing countries are:

1. **Bangladesh – 2,056,744**
2. **United States – 1,691,498**

3. **United Kingdom – 940,767**
4. **Canada – 385,938**
5. **Australia – 385,741**

Bangladesh contributes the highest number of tourists, primarily under medical tourism via land travel. Developed Western countries dominate diaspora-based tourism inflow.

5.2 Conclusion

The analysis of foreign tourist arrivals to India in 2023 reveals a highly concentrated tourism pattern influenced by geographic proximity, diaspora ties, and medical tourism demand.

South Asia emerges as the most significant regional contributor, accounting for nearly 30% of total arrivals. Developed regions such as North America and Western Europe also play a major role, contributing substantially to diaspora and leisure tourism.

The strong dominance of Indian Diaspora and Medical tourism categories indicates that India's tourism economy is supported not only by recreational travel but also by healthcare services and overseas Indian communities.

The positive skewness in data distribution shows that tourism inflow is concentrated among a limited number of high-performing countries. This suggests potential growth opportunities in underperforming regions such as Central & South America and parts of Africa.

Overall, the findings indicate that while India has strong tourism linkages with neighboring and developed countries, strategic marketing, improved connectivity, and policy interventions could help diversify and expand the global tourism base.

The use of R programming enabled systematic statistical analysis, visualization, and validation of patterns, strengthening the reliability and depth of this study.